SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE AI	**Date:**	18/08/26

Code of Conduct

*I S. Sahithya **(192472186)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S. Sahithya

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogueact sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input: Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

QUESTION 1 – Reference Resolution and Discourse Entity Linking

Given text: “Ravi met Arun at the library. He borrowed a book and later returned it.”

1. Problem Understanding

The objective is to identify the pronouns in the discourse, determine their antecedents, replace the pronouns with the correct entities, and validate the decision using discourse context. The pronouns are “He” and “it”.

2. Input and Output

Component	Details
Input	The two-sentence discourse and its referring expressions.
Referring expressions	He, it
Candidate antecedents for He	Ravi, Arun
Candidate antecedents for it	book, library, Ravi, Arun
Output	Resolved discourse with explicit entities.

3. Reference Resolution Algorithm

```
ALGORITHM Reference_Resolution(Text)
INPUT: Discourse text
OUTPUT: Resolved discourse

1.  Split Text into sentences and tokenize each sentence.
2.  Detect pronouns and referring expressions.
3.  For each pronoun P:
a. Collect candidate entities from previous discourse.
b. Apply number and gender agreement.
c. Apply recency: prefer the nearest compatible antecedent.
d. Apply semantic compatibility with the predicate/action.
e. Check discourse coherence and entity-chain consistency.
f.  Select the highest-ranked compatible antecedent.
4.  Replace P with the selected antecedent.
5.  Validate that all references are grammatically and semantically coherent.
6.  Return the resolved discourse.
END ALGORITHM
```

4. Application to the Given Text

For “He”, both Ravi and Arun are grammatically possible because both are singular male entities. The assessment text therefore does not provide an absolute grammatical distinction. Under the stated discourse-resolution strategy, recency selects Arun because Arun is the most recently introduced compatible antecedent before “He”.

For “it”, the antecedent is “a book”. This link is strongly supported by semantic compatibility: a book is a suitable object to borrow and later return, whereas the library and people are not suitable antecedents for the pronoun “it” in this action sequence.

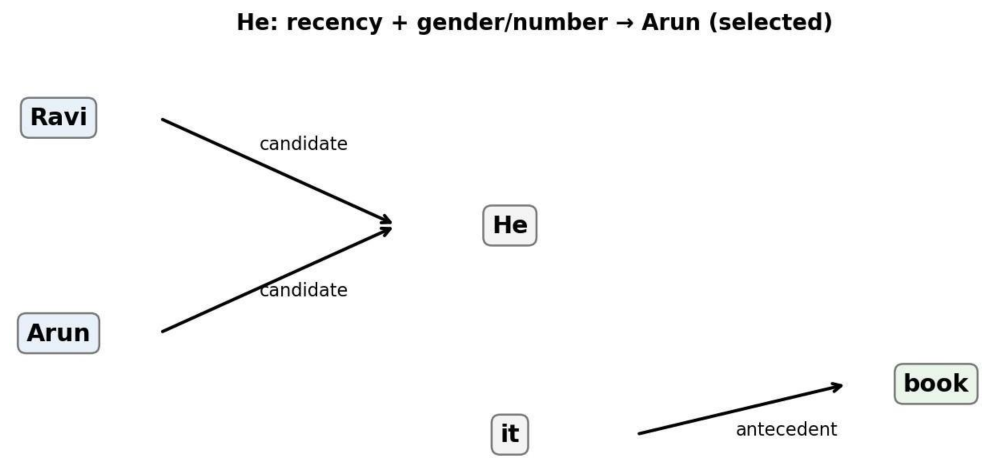

Figure 1. Entity-linking representation for the pronouns "He" and "it".

5. Resolved Discourse

"Ravi met Arun at the library. Arun borrowed a book and later returned the book."

6. Validation of the Resolution

- Gender and number: "He" is singular masculine, matching Arun.
- Recency: Arun is the most recent compatible person before "He".
- Semantic compatibility: a person can borrow a book, and a book can be returned.
- Coherence: Arun remains the participant performing both actions, producing a consistent entity chain.

Important limitation: because the source sentence does not explicitly state who borrowed the book, the resolution of "He" is a context-based inference rather than an absolutely certain fact. The algorithm should therefore assign a confidence score and flag the case if the application requires high-confidence coreference.

QUESTION 2 – Text Coherence and Discourse Structure

Given discourse: "The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

1. Problem Understanding

The objective is to identify logical relations between sentences and construct a discourse structure that explains how the sentences form a coherent unit.

2. Coherence Analysis Algorithm

```
ALGORITHM Coherence_Analysis(Discourse)
INPUT: Ordered sentences S1 ... Sn
OUTPUT: Discourse relations and structure
```

1. Split the discourse into sentences.
2. Extract events, entities, temporal markers, and connectives.
3. For each adjacent sentence pair (Si, Si+1):
a. Detect causal cues such as because, therefore, hence.
b. Detect sequence cues such as then, later, after.
c. Detect elaboration or explanation relations.
d. Compare events and identify the most plausible relation.
4. Build a discourse graph/tree using the identified relations.
5. Check whether each sentence contributes logically to the overall topic.
6. Return the discourse structure and coherence assessment.
END ALGORITHM

3. Application and Discourse Relations

Sentence	Relation	Explanation
S1: Roads were flooded after heavy rainfall.	Cause / Situation	Heavy rainfall caused flooding, establishing the situation.
S2: Schools were closed for the day.	Effect of S1	"Therefore" explicitly signals that the school closure follows from the flooding.
S3: Students attended classes online.	Consequence / Elaboration	Online classes describe how education continued after the closure.

Figure 2. Discourse structure showing the causal and subsequent relations among the three sentences.

4. Discourse Structure Representation

S1 (Flooding) → CAUSE/EFFECT → S2 (School closure) → CONSEQUENCE/ELABORATION

→ S3 (Online classes)

The structure can be viewed as a causal chain: heavy rainfall leads to flooding; flooding provides the reason for school closure; and school closure leads to the practical consequence that students attend classes online.

5. Why the Relations Maintain Coherence

- The topic remains consistent: all three sentences concern the effect of adverse weather on education.
- The connective "Therefore" explicitly links the first situation to the second event.
- The third sentence continues the situation logically rather than introducing an unrelated event.
- The sequence provides a clear progression from cause → institutional response → student response.

Therefore, the discourse is coherent because every sentence adds information that can be logically connected to the previous context.

QUESTION 3 – Dialogue Acts and Conversational Agents

Conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

1. Problem Understanding

The algorithm must identify the communicative function of each utterance and convert the conversation into a dialogue-act sequence. The required classes are Request, Question, Inform, Confirmation, or Action.

2. Intent Recognition Algorithm

```
ALGORITHM Dialogue_Act_Recognition(Conversation)
INPUT: Ordered user and agent utterances
OUTPUT: Dialogue-act label for each utterance

1. Read each utterance in conversational order.
2. Identify lexical and syntactic cues.
3. If the utterance asks the agent to perform a task, label Request.
4. Else if it asks for information, label Question.
5. Else if it supplies user information, label Inform.
6. Else if it confirms completion or acceptance, label Confirmation.
7. Else if it directly represents a system operation, label Action.
8. Use conversation context to resolve ambiguous cases.
9. Return the ordered dialogue-act sequence.
END ALGORITHM
```

3. Classification of the Given Utterances

Speaker	Utterance	Primary Act	Reason
User	Can you book a train ticket for me?	Request	The user asks the agent to perform a booking task.

Agent	Sure, where would you like to travel?	Question	The agent requests missing travel information.
User	I want to go to Chennai.	Inform	The user supplies the destination required to continue the task.
Agent	Your ticket has been booked.	Confirmation	The agent confirms that the requested booking has been completed.

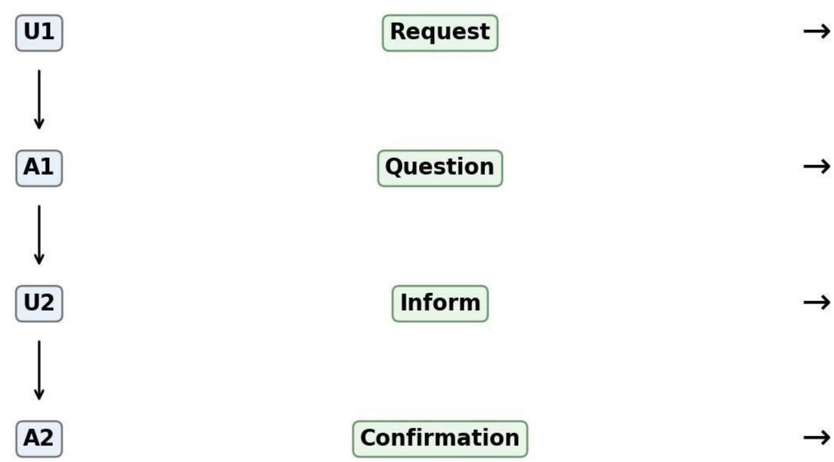

Figure 3. Dialogue-act sequence for the supplied conversation.

4. Dialogue-Act Sequence

Request → Question → Inform → Confirmation

5. Evaluation

- The Request identifies the user's main goal: booking a train ticket.
- The Question identifies missing information needed by the agent.
- The Inform supplies the destination “Chennai”, allowing the agent to proceed.
- The Confirmation closes the task by indicating that the booking has been completed.

The sequence helps the conversational agent maintain task state. It can recognize that the conversation is not a set of unrelated sentences but a structured interaction in which each act supports the next step. This improves intent interpretation, dialogue management and response selection.

Note: “Your ticket has been booked” could also be treated operationally as an Action statement because it reports a completed system action. However, among the labels specified in the question, Confirmation is the strongest primary dialogue-act classification.

QUESTION 4 – Language Generation and Surface Realization

Semantic input: Action = Buy; Agent = Student; Object = Book; Tense = Past.

1. Problem Understanding

The objective is to convert a structured semantic representation into a grammatically correct naturallanguage sentence. The generation process must perform lexical selection, sentence structuring, tense realization, and final grammatical validation.

2. NLG Pipeline Algorithm

ALGORITHM Surface_Realization(Semantic_Input)
INPUT: Action, Agent, Object, Tense
OUTPUT: Grammatically correct sentence

1. Read the semantic roles and tense.
2. Select lexical items:
   Agent → "student"
   Action → "buy"
   Object → "book"
3. Apply the specified tense:
   Past tense of "buy" → "bought".
4. Arrange the semantic roles using English SVO order.
5. Apply capitalization and determiner rules.
6. Generate the sentence.
7. Validate subject–verb agreement, tense, word order, and completeness.
8. If validation fails, revise the surface form.
9. Return the final sentence.

END ALGORITHM

Figure 4. NLG pipeline from semantic input to final surface realization.

3. Lexical Selection

The semantic role Agent is mapped to the noun "student", the action Buy is mapped to the verb "buy", and the Object is mapped to "book". Because the sentence refers to a particular book in the generated realization, the natural noun phrase is "the book".

4. Sentence Structuring and Surface Realization

English commonly uses Subject–Verb–Object order. Therefore, the semantic roles are ordered as Student + Buy + Book. The past-tense requirement changes "buy" to the irregular past form "bought".

Agent + Past Action + Object → The student bought the book.

5. Final Output and Validation

Final natural-language sentence: "The student bought the book."

Validation Check	Result
Subject present	Yes – The student
Verb present	Yes – bought
Past tense	Yes
Object present	Yes – the book
Word order	Correct SVO order
Grammar	Correct

The generated sentence is grammatically correct, semantically faithful to the input, and consistent with the requested past tense.

QUESTION 5 – Machine Translation, Interlingua and Statistical Approaches

Source sentence: “The boy is playing football.”

The task requires source analysis, Interlingua creation, candidate target-language translations, statistical scoring, final translation, and evaluation of how the combined approach improves quality. Because the question does not specify a target language, Tamil is used here as a concrete Indian-language example for demonstrating the workflow.

1. Source Sentence Analysis

Element	Value	Semantic role
Subject	boy	Agent / Player
Verb phrase	is playing	Present progressive action
Object	football	Activity object
Tense / Aspect	Present progressive	Ongoing action

2. Interlingua Representation

An Interlingua is a language-independent semantic representation. A suitable representation for this sentence is:

PLAY(agent=BOY, object=FOOTBALL, aspect=PROGRESSIVE, time=PRESENT)

This representation separates meaning from the surface grammar of English. A target-language generator can use the same semantic structure to produce an appropriate sentence.

3. Translation Workflow Algorithm

ALGORITHM Hybrid_Translation(Source_Sentence)
INPUT: Source sentence
OUTPUT: Target-language translation

1. Analyze the source sentence morphologically and syntactically.
2. Identify semantic roles, entities, tense, and aspect.
3. Convert the analysis into an Interlingua representation.
4. Generate multiple candidate target-language translations. 5. For each candidate, calculate a statistical score using learned translation probabilities and language-model likelihood.
6. Rank candidates by score.
7. Apply semantic and grammatical validation.
8. Select the highest-scoring valid candidate.
9. Return the final translation.

END ALGORITHM

Figure 5. Hybrid Interlingua + statistical translation workflow.

4. Candidate Target-Language Translations For demonstration in Tamil:

- Candidate 1: சிறுவன் கால்பந்து விளையாடிக் ககாண்டிருக்கிறான்.
- Candidate 2: அந்த சிறுவன் கால்பந்து விளையாடுகிறான்.
- Candidate 3: சிறுவன் கால்பந்து விளையாடிக் ககாண்டிருக்கின்றான்.

Candidate 1 most explicitly preserves the ongoing/progressive meaning of "is playing" and is therefore selected as the preferred candidate for this example.

5. Statistical Scoring

Candidate	Semantic fit	Grammar / fluency	Illustrative overall score
1	High	High	0.92
2	High	High	0.86
3	High	Moderate–High	0.82

The numerical values above are illustrative scoring values for demonstrating the requested ranking procedure; the supplied assessment does not provide an actual statistical model or trained probabilities. The algorithm therefore selects Candidate 1 because it best preserves the progressive aspect while remaining fluent.

6. Final Translation

Selected Tamil translation: "சிறுவன் கால்பந்F விளையாடிக் ககாண்டிருக்கிறான்."

Meaning: "The boy is playing football."

7. Evaluation of the Hybrid Approach

- Interlingua preserves the core meaning independently of the source language's surface word order.

- Explicit semantic roles reduce ambiguity about who performs the action and what the action concerns.
- Statistical scoring allows several plausible target-language realizations to be compared using learned translation and language-model preferences.
- A final semantic and grammatical validation step prevents a fluent but meaning-changing candidate from being selected.
- The combination is stronger than relying on only one component: Interlingua provides semantic control, while statistical methods provide natural and fluent target-language choices.

Conclusion

Overall, the hybrid workflow separates meaning representation from target-language realization and then uses statistical evidence to rank possible outputs. This combination can improve translation consistency, fluency and ambiguity handling, while the final validation stage protects against semantic information loss.

Assessment Requirement Alignment

The supplied Assessment Tool 2 explicitly requires necessary diagrams such as constraint graphs, coreference chains, discourse trees or predicate-logic representations where applicable, solutions satisfying the given constraints, clear evaluation, and concise evidence-supported answers. filecite turn2file0 L18-L23

All five questions from Annexure A are addressed: Reference Resolution, Text Coherence, Dialogue Acts, Language Generation/Surface Realization, and the combined Interlingua–Statistical Translation workflow. filecite turn2file0 L45-L88

The solution is also structured to match the assessment rubric, which emphasizes Problem Understanding (20%), Algorithm Design (30%), Use of Control Structures (20%), Pseudocode Structure (10%), and Completeness (20%). filecite turn2file0 L90-L164